

Product Optimization & Revenue Contribution Analysis for Afficionado Coffee Roasters

Detailed guide and project requirements for the Product Optimization & Revenue Contribution Analysis for Afficionado Coffee Roasters analysis.

Background and Context

In specialty coffee retail, not all products contribute equally to revenue.

Common challenges include:

- High-volume products with low revenue contribution
- High-priced products with low sales frequency
- Overcrowded menus that slow service and confuse customers

Retail success depends on understanding:

- Which products customers prefer
- Which products generate the most revenue

- How revenue is distributed across categories and variants

This project provides product-centric intelligence to support menu optimization and merchandising strategy

[Unified Mentor](#)

[Afficionado Coffee Roasters](#)

Problem Statement

Although transaction-level product data is available, Afficionado Coffee Roasters lacks:

- Clear visibility into product popularity vs profitability
- Category-level revenue dependency insights
- Identification of low-impact or underperforming menu items

Without structured product analytics, menu decisions are often intuition-driven, leading to inefficiencies and missed revenue opportunities.

Primary Objectives

- Identify top-selling and least-selling products
- Quantify revenue contribution by product and category
- Measure revenue concentration across the menu

Secondary Objectives

- Support menu simplification and optimization
- Identify high-impact “hero” products
- Highlight low-performing products for review or redesign

Dataset Description

Column	Description
transaction_id	Unique identifier per transaction
year	Transaction year (2025)
transaction_time	Time of transaction (HH:MM:SS)
transaction_qty	Quantity purchased
unit_price	Price per unit
store_id	Store identifier
store_location	Physical store location
Balance	Account balance
product_id	Unique identifier for each product
product_category	Broad product group
product_type	Product variant within the category
product_detail	Detailed attributes such as flavor or blend

Analytical Methodology (Step-by-Step)

Data Ingestion & Validation

- Load transaction-level data
- Validate product identifiers and prices
- Ensure quantities are positive and realistic

Revenue Computation

- Compute revenue at transaction level:
 - Revenue = transaction_qty × unit_price
- Aggregate revenue by:
 - Product
 - Product type
 - Product category

Product Popularity Analysis

- Total units sold per product
- Ranking products by sales volume
- Identify top and bottom performers

Revenue Contribution Analysis

- Total revenue per product
- Revenue share (%) of each product
- Comparison between volume rank and revenue rank

Category & Product-Type Performance

- Revenue share by category (Coffee, Tea, Chocolate)
- Product-type contribution within each category
- Dependence analysis on core categories

Revenue Concentration & Menu Balance

- Pareto (80/20) analysis

- Identify:
 - Revenue anchors (few products driving most revenue)
 - Long-tail products with minimal impact
- Evaluate menu diversification risk

Key Performance Indicators (KPIs)

KPI Name	Description
Product Revenue Contribution (%)	Share of total revenue
Product Sales Volume	Popularity indicator
Category Revenue Share	Business dependency
Revenue Concentration Ratio	Menu risk exposure
Product Efficiency Score	Revenue per SKU

Streamlit Web Application Requirements

Core Modules

- Product ranking (by volume & revenue)
- Category revenue distribution
- Popularity vs revenue scatter plots
- Product drill-down performance tables

User Capabilities

- Category and product-type filters
- Store location selector
- Top-N product slider

Deliverables and Submission

- Research paper (EDA, insights, recommendations)
- Streamlit dashboard (live analytics)
- Executive summary for government stakeholders

Conclusion

This project delivers actionable product intelligence by shifting focus from when customers buy to what they buy and what drives revenue. By uncovering product-level performance and revenue concentration patterns, Afficionado Coffee Roasters can optimize its menu, improve operational efficiency, and strengthen long-term profitability.